

OptiSight - Model Information & Technical Specifications

This document provides detailed technical specifications, architecture details, and deployment information for lightweight vision models suitable for the OptiSight agricultural robot project.

1. Moondream2

Specification	Details
Model Family	Moondream
Parameters	1.9 Billion
Architecture	Vision Transformer + Language Model
Vision Encoder	SigLIP (384×384 resolution)
Context Length	2048 tokens
Quantization	Q4_K_M (4-bit quantized)
Model Size	~1.1 GB (quantized)
Inference Speed	10-15 frames/second on mid-range GPUs
Key Strengths	Ultra-fast inference, low latency, real-time capable
Best Use Case	Sudden obstacle detection, emergency responses
Training Data	Web-scale image-text pairs
License	Apache 2.0

2. DeepSeek-VL-1.3B

Specification	Details
Model Family	DeepSeek Vision Language
Parameters	1.3 Billion
Architecture	Hybrid Vision-Language Model
Vision Encoder	SAM-based encoder with 384×384 input
Context Length	4096 tokens
Quantization	Q4_K_M (4-bit quantized)

Model Size	~800 MB (quantized)
Inference Speed	8-12 frames/second
Key Strengths	Code generation, structured output (JSON), logical reasoning
Best Use Case	Structured robot commands, API-style outputs
Training Data	Code + vision datasets, instruction-tuned
License	MIT

3. InternVL2-1B / 2B

Specification	Details
Model Family	InternVL2
Parameters	1B / 2B variants available
Architecture	Vision-Language model with dynamic resolution
Vision Encoder	InternViT-300M (448×448 native)
Context Length	8192 tokens (2B variant)
Quantization	Q4_K_M (4-bit quantized)
Model Size	~600 MB (1B) / ~1.2 GB (2B)
Inference Speed	6-10 frames/second (2B variant)
Key Strengths	Detailed scene description, excellent narration, "why" explanations
Best Use Case	Scene understanding, explaining robot decisions
Training Data	Multi-modal instruction dataset, academic focus
License	MIT

4. Qwen2-VL-2B-Instruct ★ Recommended

Specification	Details
Model Family	Qwen2-VL
Parameters	2.2 Billion
Architecture	Advanced Vision-Language Transformer
Vision Encoder	ViT with NaViT dynamic resolution (up to 1280×1280)
Context Length	32768 tokens
Quantization	Q4_K_M (4-bit quantized)
Model Size	~1.4 GB (quantized)
Inference Speed	5-8 frames/second
Key Strengths	Best CoT reasoning, video support, high-res images, agricultural object recognition
Best Use Case	Primary vision model for OptiSight, all-around performance
Training Data	Large-scale multimodal dataset with video understanding

License	Apache 2.0 / Qwen License
Special Features	Video understanding, multi-image reasoning, grounding capabilities

5. MiniCPM-V 2.6

Specification	Details
Model Family	MiniCPM-Vision
Parameters	2-3 Billion
Architecture	Compact Vision-Language with OCR focus
Vision Encoder	Enhanced ViT for text recognition (512×512)
Context Length	4096 tokens
Quantization	Q4_K_M (4-bit quantized)
Model Size	~1.8 GB (quantized)
Inference Speed	4-6 frames/second
Key Strengths	Superior OCR capabilities, small object detection, detail analysis
Best Use Case	Plant disease detection, text reading, fine-grained analysis
Training Data	OCR-enhanced datasets, document understanding
License	Apache 2.0

Deployment Requirements

Requirement	Minimum	Recommended
GPU Memory (VRAM)	4 GB	6-8 GB
System RAM	8 GB	16 GB
Storage	5 GB	10 GB (all models)
CUDA Compute	6.1+ (Pascal)	7.0+ (Volta/Turing)
CPU	4 cores	8+ cores
OS Support	Linux, Windows 10+	Ubuntu 20.04+